SHAFANDA NABIL SEMBODO

Bekasi | +62 878-2962-7598 | shafanda.nabil.s@gmail.com | Portfolio

PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer with 5+ years of experience building end-to-end ML solutions across telecom, finance, healthcare, and oil & gas industries. Experienced in data analysis, machine learning modeling, optimization, and deploying AI systems into production. Proven ability to lead data teams and collaborate cross-functionally to deliver data-driven solutions that improve operational efficiency. IBM Certified Data Science Practitioner.

KEY COMPETENCIES

Programming: Python, SQL, PySpark

Database: PostgreSQL, MongoDB

Machine Learning: Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM

MLOps & Deployment: Docker, Flask, FastAPI, MLFlow, GitHub Actions, CI/CD

Data Visualization: Tableau, Power BI, Apache Superset

Cloud & Orchestration: AWS, GCP, Apache Airflow

Other: Network Analysis, Combinatorial Optimization, Time Series Forecasting, NLP

WORK EXPERIENCE

PT. Insignia Indonesia – Jakarta

Mid. Machine Learning Engineer | *Jan 2026 – Present*

- Designed and implemented an Event-Based Root Cause Analysis (RCA) framework for PT Telkom Infrastruktur Indonesia's large-scale telecom network (~10,000+ nodes), using deterministic rule-based modeling across three network layers (access, aggregation, and core).
- Developed an end-to-end campaign planning system for Astra Digital, analyzing 25,000+ data points across 4 advertising channels (Google PMAX, TikTok, Facebook, Google Search) over a 1-year period, encompassing CPL forecasting, channel ranking, budget optimization, and LLM-powered insight generation to support data-driven marketing decisions.

PT. Pegadaian – Jakarta

Data Scientist | *Mar 2025 – Dec 2025*

- Developed route optimization models for marketing campaigns using network analysis and combinatorial optimization, improving field team coverage efficiency by 25% and reducing travel costs by 15% across 50+ regional locations.
- Built a modular M&A simulation framework enabling 5-year balance sheet and profit-loss projections, vertical/horizontal analysis, and 10+ key financial ratios to support strategic merger evaluation for executive decision-making.
- Developed a modular stress testing framework covering 6 financial components (revenue, funding, expenses, investment, financial statements, ATMR) under 3 macroeconomic scenarios, improving capital adequacy insights and risk preparedness.

Purwadhika Digital Technology School – Yogyakarta

Data Science & Machine Learning Lecturer | Jan 2023 – Sept 2024

- Taught 59 students across multiple batches with a 60–70% job placement rate post-graduation, consistently receiving student satisfaction scores above 4.2/5.0.
- Designed and delivered 58 hands-on, industry-specific modules (healthcare, banking, e-commerce, transportation) emphasizing real-world problem-solving and production-ready workflows.
- Developed 5 learning modules on Tableau and 2 modules on data storytelling, adopted as core curriculum materials.

Research Centre for Politics and Government UGM – Yogyakarta

Data Scientist | Oct 2022 – Dec 2022

- Conducted sentiment analysis on 50,000+ social media posts regarding ExxonMobil CSR programs, delivering insights used in stakeholder engagement strategy.
- Developed two end-to-end data science projects on conflict analysis in Indonesia, leveraging Twitter data and 20+ news portals to produce research reports for policy recommendations.

PT. Nanosense Instrument Indonesia – Yogyakarta

Data Scientist | May 2021 – Oct 2022

- Led the GeNose COVID-19 AI development team (5 members) through data collection, model development, and system monitoring, achieving 85–90% prediction accuracy on 10,000+ breath samples.
- Supervised data acquisition and feature engineering pipelines for Ganoderma detection and white blood cell anomaly classification projects, improving baseline model performance by 12%.

EDUCATIONAL BACKGROUND

Universitas Gadjah Mada – Yogyakarta

Bachelor of Physics Engineering | Aug 2015 – Feb 2020 | GPA: 3.58/4.00

Thesis: Designed and developed a soft sensor using a Radial Basis Function Network (RBFN) to estimate oxygen content in the debutanizer reboiler flue gas, achieving a Mean Absolute Error (MAE) of 0.216%.

LICENSES & CERTIFICATIONS

- **IBM (2021)** – *Data Science Practitioner*
- **KOMINFO (2020)** – *Data Science Course*
- **DPP UGM x Telkom (2018)** – *Big Data Analytics Course*

LEADERSHIP & SOFT SKILLS

- Led a 5-person AI development team at Nanosense for the GeNose C19 project (data collection to production deployment).
- Mentored 59 aspiring data scientists as a full-time lecturer, guiding capstone projects and career development.
- Collaborated with cross-functional teams (business, engineering, operations) at Pegadaian to translate business requirements into analytical solutions.